A $\tau_h\tau_h$ cutflow diagnostic for the Delphes HH→ $b\bar{b}\tau\tau$ sample: locating a factor ∼5 yield deficit against CMS NanoAOD

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Abstract

The tuned Delphes HH→ $b\bar{b}\tau^+\tau^-$ sample selects $\sim 5\times$ fewer $\tau_h\tau_h$ events than the CMS NanoAOD reference from the same number of generated events. A five-stage cutflow, applied identically to both samples, localises the entire deficit to a *single* stage: the FastMTT di-τ mass reconstruction, which keeps 0.111 of Delphes events against 0.994 of CMS events. Up to that stage Delphes is in fact *ahead* of CMS in cumulative efficiency. The cause is identified as a definitional mismatch rather than a detector or tuning effect: in Delphes a τ_h *is* a jet and therefore carries the AK4 jet mass ($\mathcal{O}(5\text{–}20)\,\mathrm{GeV}$), whereas the FastMTT hadronic decay prior vanishes identically unless $m_{\mathrm{vis}} \leq m_\tau = 1.777\,\mathrm{GeV}$. Roughly 89% of Delphes $\tau_h\tau_h$ pairs therefore admit no solution and return NaN. A proposed fix — an anchor-measured τ_h visible-mass tuning map — is described. This note is circulated for cross-check before the correction is adopted.

1 Motivation

The Delphes fast-simulation sample is intended to reproduce the CMS HH→ $b\bar{b}\tau\tau$ Run-3 baseline closely enough to support a phenomenological unbinned NSBI κ_λ measurement. Object-level agreement is already established: the tuned b-tagging, τ_h efficiency, energy scales and lepton scale factors are anchored to a private 2024 NanoAODv15 sample, and the Level-0/1/4 validation gates pass. The m_{HH} and m_{bb} distributions — the κ_λ-carrying observables — agree with CMS within statistics.

One discrepancy resisted every object-level fix: from 2×10^4 generated events the overlay selects ≈300 $\tau_h\tau_h$ events on Delphes against ≈1500 on CMS. Because this ratio directly caps the statistics available for NSBI training, it matters *which* stage of the selection creates it. A per-stage efficiency comparison is the minimal measurement that answers this.

2 Method

Both samples are walked through the same five stages. Definitions are matched object-for-object; where the two data tiers expose different collections, the counterpart with the same physical meaning is used.

#	stage	Delphes	CMS NanoAOD
1	gen	≥ 2 `GenJet` matched to a gen τ	≥ 2 `GenVisTau`
2	reco	≥ 2 accepted jets matched to those	≥ 2 `Tau` objects matched
3	ID	≥ 2 passing the re-tagged `TauTag`	≥ 2 passing DeepTau v2.5 VSjet Medium
4	jets	≥ 2 jets surviving the overlap removal	identical
5	$m_{\tau\tau}$	FastMTT mass finite and $> 20\,\mathrm{GeV}$	identical

Table 1: Cutflow stages. Acceptance is $p_T > 20\,\mathrm{GeV}$, $|\eta| < 2.3$ for τ candidates and $p_T > 20\,\mathrm{GeV}$, $|\eta| < 2.4$ for jets; matching and overlap removal both use $\Delta R < 0.4$.

Two choices deserve comment. First, stage 1 uses Delphes `GenJets` rather than the gen τ four-vector as the visible-τ proxy: `GenJets` are clustered after the neutrino filter and are therefore already the visible object, matching the definition of `GenVisTau`. Binning the Delphes side by the *full* gen-τ p_T would compare different quantities, since a hadronic τ carries only $\sim$65% of its momentum visibly. Second, stage 4 applies a symmetric overlap removal: the di-τ pair is selected first, then jets within $\Delta R < 0.4$ of either selected τ are removed on *both* sides. Without this, the Delphes side removes τ-tagged jets from the b pool while the CMS side does not (there τ_h live in a separate `Tau` collection but their jets remain in `Jet`), which previously produced a spurious ΔR_{bb} discrepancy.

3 Results

Tables 2–4 give the cutflow at three κ_λ points, from 2×10^4 generated events each, with the tuning maps applied. $\varepsilon_{\text{step}}$ is the conditional efficiency of each stage, ε_{tot} the cumulative one, and $\Delta\varepsilon$ the Delphes/CMS ratio of $\varepsilon_{\text{step}}$; values far from unity mark the stage responsible.

	Delphes			CMS NanoAOD			
stage	N	$\varepsilon_{\text{step}}$	ε_{tot}	N	$\varepsilon_{\text{step}}$	ε_{tot}	$\Delta\varepsilon$
0 events read	20000	—	1.0000	20000	—	1.0000	
1 gen visible τ_h	9859	0.493	0.4929	4657	0.233	0.2329	2.12
2 reco τ_h object	6330	0.642	0.3165	3965	0.851	0.1983	0.75
3 τ_h ID	2268	0.358	0.1134	1465	0.369	0.0732	0.97
4 cleaned jets	1961	0.865	0.0980	1267	0.865	0.0634	1.00
5 FastMTT $m_{\tau\tau}$	**217**	**0.111**	0.0109	1259	0.994	0.0630	**0.11**

Table 2: Cutflow at $\kappa_\lambda = 0$. Final CMS/Delphes yield ratio 5.80.

	Delphes			CMS NanoAOD			
stage	N	$\varepsilon_{\text{step}}$	ε_{tot}	N	$\varepsilon_{\text{step}}$	ε_{tot}	$\Delta\varepsilon$
0 events read	20000	—	1.0000	20000	—	1.0000	
1 gen visible τ_h	9987	0.499	0.4994	4708	0.235	0.2354	2.12
2 reco τ_h object	6339	0.635	0.3170	4027	0.855	0.2014	0.74
3 τ_h ID	2348	0.370	0.1174	1554	0.386	0.0777	0.96
4 cleaned jets	2030	0.865	0.1015	1375	0.885	0.0688	0.98
5 FastMTT $m_{\tau\tau}$	**243**	**0.120**	0.0121	1362	0.991	0.0681	**0.12**

Table 3: Cutflow at $\kappa_\lambda = 1$ (SM). Final CMS/Delphes yield ratio 5.60.

	Delphes			CMS NanoAOD			
stage	N	$\varepsilon_{\text{step}}$	ε_{tot}	N	$\varepsilon_{\text{step}}$	ε_{tot}	$\Delta\varepsilon$
0 events read	20000	—	1.0000	20000	—	1.0000	
1 gen visible τ_h	8863	0.443	0.4431	4257	0.213	0.2129	2.08
2 reco τ_h object	6101	0.688	0.3050	3487	0.819	0.1744	0.84
3 τ_h ID	2064	0.338	0.1032	1083	0.311	0.0541	1.09
4 cleaned jets	1758	0.852	0.0879	932	0.861	0.0466	0.99
5 FastMTT $m_{\tau\tau}$	**195**	**0.111**	0.0097	928	0.996	0.0464	**0.11**

Table 4: Cutflow at $\kappa_\lambda = 5$. Final CMS/Delphes yield ratio 4.76.

Reading the tables. Stages 3 and 4 agree between the two samples to within a few percent at all three κ_λ points ($\Delta\varepsilon = 0.96$–1.09 and 0.98–1.00): the τ_h identification and the jet selection are *not* the problem. Stage 5 is catastrophic and consistent across κ_λ: Delphes retains 11–12%

where CMS retains 99%. Crucially, the cumulative efficiency through stage 4 is *higher* on Delphes (0.098 vs 0.063 at $\kappa_\lambda = 1$); the entire deficit, and more, is generated at stage 5 alone.

Known artefacts in stages 1–2. The stage-1 ratio $\Delta\varepsilon \simeq 2.1$ is an artefact of the diagnostic itself, not of the simulation. The Delphes `GenJet` proxy cannot distinguish a hadronic from a leptonic τ decay — a $\tau \to \ell\nu\bar{\nu}$ also produces a `GenJet` — so the Delphes denominator is inflated by roughly $1/\mathcal{B}(\tau_h\tau_h)$. Stage 2 ($\Delta\varepsilon \simeq 0.75$) is the mirror image: those leptonic-τ events fail to produce a τ-jet. The two largely cancel and neither contributes to the final deficit. Removing this artefact requires the gen decay chain, which the current reader does not retain.

4 Cause

In Delphes a τ_h candidate *is* a jet, so the object handed to the di-τ mass estimator carries the AK4 jet mass — underlying event and additional particles clustered within $R = 0.4$ — typically 5–20 GeV. A reconstructed CMS τ_h instead carries its decay-mode visible mass, bounded above by $m_\tau = 1.777\,\text{GeV}$ ($\pi^\pm$: 0.14; ρ: 0.77; a_1: 1.26 GeV).

The covariance-free FastMTT estimator maximises, per event, a likelihood over the visible energy fractions $x_1, x_2 \in (0, 1]$,

$$\mathcal{L}(x_1, x_2) \;=\; \mathcal{L}_{\text{MET}}(x_1, x_2) \prod_{i=1,2} \mathcal{L}_{\text{dec}}(x_i)\,, \qquad m_{\tau\tau} \;=\; \frac{m_{\text{vis}}}{\sqrt{x_1 x_2}}\,, \tag{1}$$

where the hadronic decay prior is the phase-space density

$$\mathcal{L}_{\text{dec}}(x) \;=\; \begin{cases} 1, & x_{\min} \le x \le 1\,, \\ 0, & \text{otherwise}\,, \end{cases} \qquad x_{\min} \;\equiv\; \left(\frac{m_{\text{vis}}}{m_\tau}\right)^2 . \tag{2}$$

If $m_{\text{vis}} > m_\tau$ then $x_{\min} > 1$ and the admissible interval in Eq. (2) is *empty*: the prior vanishes for every x, the likelihood is identically zero, and the estimator returns NaN. Table 5 shows the effect on a 40-point scan.

m_{vis} [GeV]	$x_{\min}$	valid grid points	
0.14 ($\pi^\pm$)	0.006	40/40	
0.77 (ρ)	0.188	33/40	CMS τ_h
1.26 (a_1)	0.503	20/40	
1.78 ($= m_\tau$)	1.000	1/40	
3.0	2.850	0/40	
8.0	20.27	0/40	typical Delphes τ-jet
15.0	71.25	0/40	

Table 5: Number of admissible grid points in the FastMTT x scan as a function of the leg visible mass. Above m_τ no solution exists.

The 11% of Delphes events that do survive are those rare pairs in which *both* τ-jets happen to have a mass below m_τ — a biased, unusually narrow subset. This also offers a natural explanation for two residual shape discrepancies seen in the feature overlays, namely a $m_{\tau\tau}$ peak shifted $\sim$10 GeV high and a $\Delta R_{\tau\tau}$ distribution displaced to larger values, and for the low acceptance×efficiency (0.082) reported by the Level-4 m_{HH} gate.

5 Proposed correction

The Delphes τ_h four-vector should carry a physical τ visible mass. The jet mass is a clustering artefact, not a property of the τ, and since $m_{\tau\tau} = m_{\text{vis}}/\sqrt{x_1 x_2}$ it sets the $m_{\tau\tau}$ *scale* as well as determining whether a solution exists at all.

The correction adopted follows the existing tuning pattern used for the energy scales: the median τ_h visible mass is measured on the CMS anchor as a function of p_T (from `Tau` objects passing DeepTau Medium), stored as a tuning map, and applied downstream to τ-tagged Delphes jets. It requires no change to the Delphes card and no re-production: the existing samples are corrected in place at analysis level. The implementation applies the map after the energy-scale correction (so the lookup uses the corrected p_T and the assigned mass is not subsequently rescaled), caps the assigned mass strictly below m_τ, and leaves all non-τ jets untouched, so m_{bb} and the b-side observables are unaffected.

Known limitation. The map stores the per-p_T *median* mass and assigns it deterministically. The true visible-mass distribution is essentially trimodal in the decay mode, so the spread is not reproduced. The expectation is that this affects the $m_{\tau\tau}$ resolution rather than its scale; this is one of the points on which cross-check is sought.

6 Points for cross-check

1. **Is the diagnosis accepted?** The claim is that a single definitional mismatch — jet mass versus τ visible mass — accounts for the entire $\tau_h\tau_h$ yield deficit, and that no detector-level or tuning effect need be invoked.
2. **Is the median-mass assignment adequate**, or should the decay-mode composition be sampled so the visible-mass spread is reproduced? Does collapsing to a median bias the $m_{\tau\tau}$ scale as well as the resolution?
3. **Should the τ_h four-momentum be corrected further?** Only the mass is addressed here; the τ-jet direction and energy already receive an energy-scale correction, but the jet axis may differ from the HPS τ_h axis.
4. **Stage-1/2 artefact.** Is it worth retaining the gen decay chain in the reader so hadronic and leptonic τ decays can be separated, making stages 1–2 directly comparable?
5. **Residual shape differences.** If the correction does not fully resolve the $m_{\tau\tau}$ and $\Delta R_{\tau\tau}$ shifts, the next candidate is the MET resolution, which the tuning lens still reports as requiring adjustment (Delphes MET is *cleaner* than the CMS anchor) and which enters FastMTT directly.

Reproducing

The diagnostic is `scripts/tauh_cutflow.py` in the `delphes-pipeline` repository:

```
python scripts/tauh_cutflow.py --config config.v1.yml \
    --delphes-dir /ceph/jpan/gen-delphes \
    --nano-dir    /ceph/jpan/cms_nanoaod_2024_hh2b2tau \
    --max-events 20000 --out outputs/validation_v1/tauh_cutflow.md
```

Unit tests inject a loss at one known stage at a time and assert the cutflow attributes it to that stage and no other.